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COMMERCIAL ITEM DESCRIPTION
SURVEY INSTRUMENT, AUTOMATED, INTEGRATED
(AISI)

The General Services Administration has authorized the use of this commercial item description for all federal agencies. The original issue of this CID replaced MIL-S-0053085A(ME).

1 SCOPE. This specification covers manually operated and robotic, portable, medium-range (two-kilometer (km) and seven km), electro-optical distance and angle measuring equipment capable of second order, class II surveys. Distances are measured with an Electronic Distance Meter (EDM), using diode-emitted, modulated, infrared light waves; and angles determined by electronic measurement. The AISI measuring unit will be of solid-state construction and complete with accessories for unsupported operation.

2 CLASSIFICATIONS. AISIs will be of the following classifications, as specified (see 7.2).

2.1 Type. The AISIs will be of the following types:

Type I – With EDM of 7 km capability.

Type II – With EDM of 2 km capability.

2.2 Class. The AISIs will be of the following classes:

Class A – With winterization kit.

Class B – Without winterization kit.

2.3 Style. The AISIs will be of the following styles:

Style 1 – Without robotic capability.

Style 2 – With robotic capability.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: US Army Engineer Research and Development Center, Topographic Engineering Center, 7701 Telegraph Road, ATTN: CEERD-TS-T, Alexandria, VA 22315-3864.

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3 SALIENT CHARACTERISTICS

3.1 Description. The automated integrated survey instrument, hereinafter referred to as AISI, shall combine into a single, compact, field survey, total station system, the three major subassemblies (modular or integrated) listed below:

a. An Electronic Theodolite (ELT), an auto-reading/recording digital theodolite with electronic display/control panel(s);
b. An Electronic Distance Meter (EDM), a low power, miniaturized distance measuring device with a range and accuracy compatible with the ELT;
c. The Data Recorder Unit (DRU), an automatic data recording/storage/processing terminal, ruggedized for field use with the ELT and EDM, interoperable with any microprocessor through an RS232 interface or using industry standard (removable) data storage media and provide high-speed data transfer.

The AISI shall be mountable for operation on the standard tripod specified in A-A-52298, using the universal tribrach specified therein, or equivalent. The AISI shall also include transport carry case(s); retro-directive prisms; tribrachs; necessary accessories and tools, with cases; and survey system software programs (see 7.2) for computation, plotting and printing. The system shall be provided with (2) rechargeable battery packs; a battery charger; an auxiliary external power cable, not less than 6 meters in length; or a comparable, self-contained, electrical power subsystem. When specified (see 7.2), robotic capability (see 3.9.5), a winterization kit, thermometer, altimeter-barometer, meteorological correction data, a solar filter, tripods, tripod bases, and a backpack shall also be included. The AISI shall electronically measure values for subtended horizontal and vertical angles; slope distances up to a minimum range of 7 km or 2 km, for the type I or type II instrument, respectively; and automatically record these data to a data storage/processing module for on-site reduction, or transfer via a serial or other industry standard interface to an external microprocessor. The robotic capability shall permit one-man survey operations up to a distance of not less than 1 kilometer. The AISI shall be capable of full daytime and nighttime operation. Additionally, the AISI shall be able to cue the layout of horizontal and vertical earthmoving operations when provided with appropriate input data. The AISI should minimize the use of cabling and allow for new features and technologies, as they become available.

3.2 Safety. The AISI shall present no uncontrolled hazards to users, maintenance personnel, or bystanders. The AISI shall meet either the power and irradiance requirements for a class I device listed in TB MED 524, or be less than the irradiance criteria for a class I device and less than five times the power limitations. In all cases, the AISI shall be eye safe.

3.3 Maintainability.

3.3.1 Maintenance support. Assemblages or support elements such as repair parts, special tools, and test equipment shall be adequate to perform the intended functions of assisting in or conducting maintenance operations.

3.3.2 Ease of maintenance. AISI construction shall provide ease of access to power supplies and fuses (or reset buttons) such that tools are not required to perform user maintenance. A self-test system with a display to verify satisfactory operating capability prior to initial measurement shall be provided.

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3.4 Physical characteristics.

3.4.1 Portability. The AISI shall be furnished in soldier portable packages.

3.4.2 Weight. The basic equipment (ELT, EDM, DRU, tribrach, and power pack) weight shall not be greater than 11.35 kg. The transport weight (basic equipment and transport carry case) shall not be greater than 18.18 kg.

3.4.3 Electronic theodolite (ELT). The horizontal and vertical angle measuring component of the AISI, the ELT, shall automatically display the observed angle at a precision and repeatability required to satisfy 2nd order, class II accuracy of National Oceanic and Atmospheric Administration (NOAA) Publication (Standards and Specifications for Geodetic Control Networks, September 1984).

3.4.3.1 Alidade assembly. The alidade assembly shall be provided with a self-leveling device, and a means for attaching to the tribrach.

3.4.3.1.1 Telescope. The telescope shall have an aiming device, sunshade, and a focusing range of 1.7 meters to infinity. The telescope shall feature not less than a 30x magnification, and shall produce an erect image.

3.4.3.1.1.1 Reticle. The reticle shall consist of a pattern of opaque lines similar to Figure 4. The stadia ratio shall be 1/100. The horizontal and vertical collimation adjustment of the reticle shall be positive and easily accessible. The horizontal and vertical collimation errors of the lines of sight shall be not greater than 0° 0' 6".

3.4.3.1.1.2 Telescope eyepiece. The eyepiece shall be equipped with a diopter scale marked with plus and minus units from zero, and shall have an index mark for referencing the setting.

3.4.3.1.2 Standards. The horizontal axis of the telescope shall be so nearly perpendicular to the vertical axis of the AISI, that the error due to the difference in height of the standards shall not be greater than 0° 0' 12"; or shall be compensated for by onboard software.

3.4.3.1.3 Circles.

3.4.3.1.3.1 Horizontal and vertical circle. The horizontal and vertical circles shall provide a means of automated correction for circle eccentricity. The resolution of the horizontal circle readings shall not be greater than 0° 0' 0.5". The time to initial display shall not exceed one second. Inclination of the vertical axis shall be corrected automatically. Units of angular measurement shall be user selectable in either sexagesimal, decimal, or mil readout. A means to correct for the influences of mis-levelment in the vertical and trunion axes shall be incorporated; it may be provided as a discrete compensator.

3.4.3.1.3.2 Setting circles. A horizontal and vertical circle setting capability shall be furnished for setting the circles in various positions of an observation program. The circles shall be provided with a means of selecting the units of angular measure so that they correspond to those selected on the microprocessor.

3.4.3.1.4 Microprocessor display/control panel. At least one display/control panel shall be supplied on the ELT/EDM; and one shall be provided on, or transferable to, the prism pole for style 2 units. The prism pole panel shall be identical, or sufficiently similar, to that on the ELT/EDM, to provide identical functionality in remote control of the unit without revised

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operating characteristics. Additional controls for remote operation shall not assume placement of the original ELT/EDM controls. It shall be provided with a subsystem for uninterruptible communication with the ELT/EDM (see 3.9.5.2).

3.4.3.1.4.1. Keyboard. Panel dials, displays, knobs and switches shall be functionally grouped for simple operation. Each shall be marked to identify its functional name, symbol, and operating instruction reference. The keyboard shall be weatherproof and shall be capable of initiating the microprocessor functions listed in 3.4.3.1.4.4.

3.4.3.1.4.2 Displays. Each display shall depict operational commands, data entry commands, self-test status, and data input, as a minimum. Displays shall be legible in any ambient light.

3.4.3.1.4.3 Illumination. The electronic control panel shall be sufficiently illuminated for night operation. All lighting external to the unit shall be a blue-green color between 400-700 nanometers in wavelength. The intensity of all externally visible light produced shall be adjustable. For style 1, no light shall be visible beyond a 50-meter circle, with the instrument as center point, under moonless and starless night conditions. In style 2, the tracking/positioning lights shall have an on/off control.

3.4.3.1.4.4 Microprocessor. The AISI microprocessor shall be a fully programmable operating environment; having nonvolatile memory of adequate capacity to store all instrument constants and operating parameters, and enough random access program memory to perform required data reductions. Upon start-up, the microprocessor shall initiate and control a system self-test that shall include power source check, EDM operational check, and display test sequence. The microprocessor also shall be capable of executing the following listed functions automatically or from the control panel keyboard input:

OPERATIONAL COMMANDS
- Power on/off
- Illumination on/off
- System self-test
- Clear entries/commands
- Select/display units of angular/distance measure
- Select precision of angular display readings
- Set horizontal circle orientation
- Measure/display/record horizontal angle or direction
- Measure/display/record vertical or zenith angle
- Measure/display/record horizontal collimation error
- Measure/display/record inclination of standing axis
- Level standing axis
- Measure/display/record slope or horizontal distance
- Measure/display/record height differences or elevations
- Display coordinates
- Display stake-out direction/distance
- Display error/system messages

DATA ENTRY COMMANDS
- Enter vertical circle index error
- Enter horizontal collimation error

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Enter stakeout data
Enter initializing horizontal/vertical datum
Enter height of instrument/target
Enter instrument/target constants
Enter scale corrections
Enter atmospheric data/corrections
Enter system codes (stationing, record)
Enter system format parameters (bits per second (bps), parity etc.) for data transmissions
Enter instrument/target eccentricity.

3.4.3.1.5 Axes motion controls. Movement through the vertical tilt limits shall be a minimum of ± 45°. Horizontal and vertical motion controls shall provide for fine and coarse adjustment.

3.4.3.1.6 Plummet. The plummet shall be integral with the alidade or shall be attached to the tribrach and be adjustable. If an optical system is used, the reticle pattern shall divide the field of vision into four 90° 0' 0" quadrants. The system shall be useable throughout the operating range from tripod height to not less than 30.5 meters. The plummet shall be coincident with the vertical axis of the AISI. The plummet shall be capable of being used in daytime and nighttime conditions.

3.4.3.1.7 Levels. The instrument shall be equipped with a leveling capability sufficient to allow it to be setup and operated at required accuracies under any of the conditions described herein.

3.4.3.1.8 Tribach. The tribrach shall consist of the leveling screws, base plate, and a device for locking the alidade assembly. For compatibility with existing equipment, the base plate shall have at its center a 5/8-inch nut of the unified coarse thread series, class 2B, 11 threads per inch (5/8-11 UNC-2B). The leveling screws shall afford leveling of the theodolite when the base plate is 6° 0' 0" out of the level. The design of the tribrach shall be such that removing and returning the alidade shall not cause a change in the centering of the alidade greater than 0.05 mm. If the optical plummet is integral to tribrach see 3.4.3.1.6.

3.4.3.2 Power. The ELT/EDM shall operate from an internal or external battery or other self-contained power source. When batteries are provided, NiMH cells are preferable. The AISI shall contain built-in protection against reverse polarity from the power source, including a mount or compartment that precludes improper assembly of the power pack. The AISI winterization kit heating elements may operate from an external source (e.g., vehicle battery).

3.4.3.2.1 System power. The ELT/EDM shall be provided with two power packs; each capable of supporting a minimum of 500 angle and distance measurements at 20 ºC, and interchangeable without loss of data. Power packs shall be rechargeable (see 3.4.3.2.4), spillproof, non-corrosive, and have no detrimental liquid or gas seepage, regardless of operation or storage orientation.

3.4.3.2.2 Remote power. The remote control panel and transmitter/receiver on the prism pole of the style 2 instrument shall be provided with the same, or similar, power pack(s) as those on the ELT/EDM (see 3.4.3.2.1).

3.4.3.2.3 Auxiliary power. The ELT/EDM shall also operate from external 12-volt DC batteries, providing the same operating capability as with the internal power source. An external power cable shall be provided with the system that shall not be less than 6m in length and shall terminate in automotive type battery clamps, identified as positive and negative.

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3.4.3.2.4 Charger. The power packs shall be rechargeable using a 110/220 volt, 50/60 Hz charger(s) that shall be supplied with the system. The charger shall be capable of concurrently charging at least two power packs or a built-in charger shall be furnished with each power pack. The power pack shall not be damaged nor lose capacity when charged continuously from the charger supplied with the system.

3.4.4 Electronic distance meter (EDM). Distance measurements shall be acquired in a normal mode (individual measurements manually cued and displayed), tacking mode (continuously measured and displayed), or distance-averaging mode. The keyboard controls shall allow for the entry of the following commands: Power/illumination on-off; measure and display distance; activate tracking mode; stop tracking mode; self-test (signal strength, power level); data entry (prism constant, scale correction); and the switching to/from remote operation, for style 2 units. The units of distance measurement and data entry shall be keyboard selectable in either the Metric or English system. The keyboard shall provide the above functions and in addition, extend the capabilities of the ELT/EDM combination by providing for remote EDM control, horizontal distance display, height difference, differential ground point stakeout, coordinate differences, and atmospheric corrections. Automatic correction for atmospheric conditions is desirable. The system shall also accept data input from the software (see 3.10).

3.4.5 Data recording unit (DRU). This component of the AISI shall automatically acquire and store data from the ELT/EDM, when required, download the data to an external data processing unit or storage medium. It shall be programmable to perform data storage, transfer, and customized computing tasks. The programming language shall be user friendly. Prepared programs shall be supplied, as specified (see 7.2), to accomplish common field routines such as coordinate geometry calculations, stakeout, precise leveling, and construction alignment. A means of reference to frequently entered field data codes, DRU commands, or program instructions shall be attachable to the ELT or DRU for user convenience. Non-volatile memory capacity shall be adequate to store data transfer parameters (bps, parity, etc.) and field station data (horizontal and vertical angle, horizontal distance, and station designation/information code) for not less than 500 survey stations. The power source necessary to maintain memory non-volatility shall have a minimum supporting life of not less than 12 months.

3.4.5.1 Data structure. In addition to data manipulation, the DRU shall be capable of recording, accessing, editing, and executing program files. Allocation and organization shall allow separate storage and retrieval of data files containing station identification information, horizontal angle, horizontal distance and vertical angle parameters for a minimum of 500 ground control points, per megabyte (Mb). The unit shall be provided with not less than 1Mb of storage, preferably removable. The design, size, and number of both data and program files shall be sufficient to support this field survey data collection, encoding, and program commands. The DRU shall be capable of data transfer using standard interfaces such as RS232 or PC card. The transfer rate shall be a selectable transmission at not less than 2400 bps. Parity, character format, protocol, and terminal emulation also shall be selectable.

3.4.5.2 Display. Displays shall be legible in any ambient light as well as indirect or artificial light. Multiple or single displays shall be capable of the presentation of all prompts, program instructions and data. Display format shall be user-selectable, without interchanging modules or the need to re-program the device.

3.4.5.3 Power. The DRU shall be powered by a rechargeable power supply. Operating power loss shall not affect data stored in the non-volatile memory and shall be preceded by a display warning.

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3.4.6 Reflector set. Two retro-reflector prism sets shall be provided as a component of the AISI. Each set shall consist of the following.

a. Sufficient number of prisms to satisfy maximum range capability for the AISI (see 3.9.4.2).
b. Single and multi-prism holders to accommodate all prisms provided.
c. Prism adapter to allow mounting of a combination of prism holders, when specified (see 7.2).
d. Mounting adapter(s) for mounting both the prism holders and adapter in the universal tribrach specified in 3.4.3.1.8. The mounting adapter shall permit prisms to be rotated through 360 degrees of arc, about the vertical axis, without disturbing either the tribrach or plumb of the assembly over the survey mark. The adapter shall include a means to plumb over the survey mark. In addition, a means shall be provided to restrict movement of the prisms at any point after rotation.
e. Transport carry case or cases conforming to 3.4.7.

3.4.7 Transport carry cases. Each carry case (housing the electro-optical package, accessories and tools) shall be of a rigid construction with a removable inner liner, molded to accept the various AISI accessories. The cases shall withstand operating, transport, and storage usage and the contents shall be adequately blocked and padded to prevent damage. The cases shall prevent any damage to the AISI when dropped from heights up to 91 cm. The cases shall also protect the instrument against damage from rainfall, shock, and vibration as specified herein. Each case shall contain a desiccant. Each case shall open in such a way that the contents will not fall out when the case is opened. The cases shall be provided with heavy-duty locking catches or means for padlocking. Except the backpack, tripod, and tripod base, all items listed in 3.4.9 shall be provided in carry cases. The cases shall be no larger than 63.5 cm x 30.5 cm x 25.4 cm. Each carry case shall weigh no more than 11.35 kg and labeled with a loading plan (see 3.5).

3.4.8 Accessories and tools. The contractor may provide additional items as required to perform user maintenance and field adjustments. Accessories and tools shall be provided with a separate case, similar in construction to that indicated in 3.4.7, or packed with major components in their respective transport carry cases (the pole, and plummet may be packed in its own case). The following shall be the minimum furnished with each AISI as accessories and tools:

(List on next page)

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Item	Quantity
Spare Fuses (when used)	4 each
Brush, lens	1 each
Tissues, lens	1 set
Users manual	1 each
Covers, lens, ELT, (both ends)	1 set
Cover, lens, EDM	1 set
Cover, instrument, waterproof	1 each
Pole, prism; 2.44 meters (adjustable), with level	2 each

3.4.9 Optional equipment. When specified in ordering data (see 7.2), the following equipment will be provided with the AISI:

Item	Quantity
Backpack (see 3.4.9.1)	2
Winterization kit (see 3.4.9.2)	1
Tripod (see 3.4.9.3)	3
Tripod base (see 3.4.9.3.1)	3
Shield, Solar Radiation (see 3.4.9.4)	1
Thermometer, centigrade (commercial standard)	1
Altimeter/barometer, (mm Hg graduation) commercial standard, with meteorological correction charts	2
Adapter 110/220v, 50/60 Hz (for use with battery charger.)	1

3.4.9.1 Backpack. When specified (see 7.2), a means for transporting AISI by backpacking shall be provided. The backpack shall be water-repellent and a subdued color or camouflaged. The backpack shall be easy to use and shall be comfortable to wear for male and female personnel ranging anthropometically from the 5th to 95th percentile.

3.4.9.2 Winterization kit. When specified (see 7.2), a winterization kit shall be furnished for use with the AISI at temperatures below -20 ºC. The winterization kit shall include covers for all control knobs and leveling screws (unless plastic or plastic coated), and shields for the telescope and the plummet eyepieces. The kit shall be contained in a pliant, water-repellent carrying case. The kit shall be capable of heating AISI to required operating temperature from -32 ºC within 25 minutes. An auxiliary power source (see 3.4.3.2.2) may be used. An option to procure a unit certified to be capable of operation in sub-zero temperatures, in lieu of a separate kit (Class A), may be exercised by the procurement officer (see 7.2).

3.4.9.3 Tripod. When specified (see 7.2), tripods conforming to A-A-52298 shall be furnished.

3.4.9.3.1 Tripod base. When specified (see 7.2), a base suitable for use with the A-A-52298 tripod shall be furnished (see 7.6). The base is a device that stabilizes the tripod on hard surfaces (roadways, tarmacs, etc.), which prevent penetration of the tripod leg tines into the surface.

3.4.9.4 Solar shield. When specified (see 7.2), a solar filter that will permit observations of the sun shall be provided. The filter shall not interfere with operation of the instrument when not in use, if integral; and provided with its own protected storage, if detachable.

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3.5 Identification marking. A loading plan depicting intended location of stored components shall be permanently attached to the top inside of the transport carry case cover, or the inserts of the transport case shall have cut-outs that clearly indicate where the item should be placed.

3.6 Interchangeability. All components from the same manufacturer having the same system part number or function shall be functionally and dimensionally interchangeable.

3.7 Treatment. Exposed, exterior surfaces of the AISI, its cases, and accessories, subject to deterioration from the elements, shall be painted (or otherwise protected) in accordance with the manufacturer's best commercial practices.

3.8 Lubrication. All portions of the AISI requiring it shall be lubricated prior to assembly. The lubrication shall permit the AISI to operate throughout the specified temperature ranges.

3.9 Performance.

3.9.1 Calibration. Each AISI shall be calibrated so that there is no instrument constant, retrodirective prism offsets, or other detrimental effects that cannot be corrected prior to distance and angle measurement. If such constants, offsets, or effects do exist, they shall be recorded on the applicable hardware in a conspicuous place to facilitate presetting the AISI. Means shall be provided for adjusting the AISI to the proper constant without use of special tools.

3.9.2 Measurement time. After setup and actuation of the AISI's off-on power switch to the ON position, the time required for an initial distance and angular reading in ambient temperatures from -20 to 40 ºC shall be not greater than 4 minutes. Subsequent distance and angular readings shall be accomplished in not greater than 6 seconds. The same time limitations shall apply to the AISI after having been warmed from -32 to -20 ºC by the winterization kit as specified in 3.4.9.2, when applicable. The instrument, in any mode of operation, shall be capable of making and recording distance and angular measurements in not greater than 6 seconds (0.03 mil).

3.9.3 Electronic theodolite.

3.9.3.1 Measurement accuracy. The measurement of horizontal and vertical angles shall yield a Standard Deviation of mean pointings between direct and reverse sightings of 0° 0' 0.8" or less.

3.9.3.2 Data processing, storage, and transfer. Field measurement data shall be capable of being automatically and manually manipulated, as defined in 3.4.3.1.4.4, without loss of information or identification. For style 2 units, this capability shall also be implemented at the control/display on the prism pole.

3.9.4 Electronic distance meter.

3.9.4.1 Measurement accuracy. The maximum allowable mean-square error for a mission of 10 observations, at an ambient temperature range of -5 to 50 ºC, shall not exceed ±5mm, +5ppm in standard mode and ±10 mm, +5ppm in tracking mode. (Note: Temperatures shall be accurate to ±1 ºC and pressures to ±5mm of mercury.).

3.9.4.2 Range. Atmospheric conditions and number of prisms used as targets determine the range capabilities of the AISI. The AISI shall measure distances down to 2 meters or less. The type I AISI shall measure distances to 7 km or more and type II AISI shall measure distances to 2 km or more. The type I, style 2 AISI shall measure distances to 1 km or more in robotic mode and to 7 km or more with the robotic mode turned off. The type II, style 2 AISI shall measure

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distances to 0.7 km or more in robotic mode and to 2 km or more with the robotic mode turned off. These distances shall be achieved under average atmospheric conditions; defined as 15-20 km visibility, light haze, moderate sun, and light heat radiation. All measurements shall meet the accuracy specified in 3.9.4.1.

3.9.5 Robotics. The style 2 AISI shall be a self-contained robotic unit capable of performing all elements of measurement defined herein, remote to ELT/EDM. The communication subsystem (telemetry unit) shall be capable of governing the ELT/EDM from the target rod, without cabling, permitting survey operations by a single individual. The system shall be capable of making and recording automatic measurements, at predetermined distance or time intervals, without specific initiation by the user.

3.9.5.1 Tracking. The system shall be provided with a tracking module capable of repositioning the instrument to track and measure to a roving target (prism(s). The subsystem shall be capable of positioning the instrument in 0º 0' 1"-increments, at speeds to 30º per second, minimum; and of locating or relocating the target in not greater than 45 seconds. A separate module may integrate with the optical tracking to achieve this. Prisms shall meet the minimum requirements specified in 3.4.6, and be distinguishable by the ELT/EDM, from other reflectors/reflecting surfaces that may be present in the survey area. A powerless, retro-reflecting prism is preferable. The prism (or array) shall have a 360° target surface for mapping purposes; non-360° targets are permitted for traversing applications. Auto-positioning accuracy shall be within ± 4mm/100m. The range of the 360° target shall be 700 m in robotic mode.

3.9.5.1.1 Search capability. The unit shall be able to recover from a, three-second minimum, momentary loss of signal; and shall be capable of re-targeting the prism. The system shall automatically re-target, within a definable search window, and shall accept instructions from the remote user to direct it in locating the prism. The unit shall be provided with an optical cueing system to direct the remote user to a predetermined survey location; and instructions on the remote display shall be unambiguous.

3.10 AISI system software. The system software shall be a fully integrated, three-dimensional, ground modeling, design, and drafting system. The system shall include modules/applications for automated data collection, traverse balancing, surface modeling, contour mapping, coordinate geometry, earthwork volumes by average-end-area or true prismoidal calculations, and three-dimensional projections. The system software may communicate directly with the surveying instrument for downloading of raw survey data and uploading of design stakeout. It is desirable that it also includes field note reduction, roadway design and drainage design modules, and an automatic determination of scale factors capability. The software shall be executable from a 32-bit Windows® environment and shall run on IBM™ and IBM-compatible hardware. Other state of the art operating environments may be permissible, providing the output file formats are Windows compatible. The software may contain separate applications to perform diverse functions, such as drafting and design calculations. However, maximum integration into one package is highly desirable. Software application(s) shall be in the two areas of topographic (control) survey and construction survey. In both areas, the software shall be required to transfer data between recorder and processor, and to peripheral equipment such as a printer or plotter. Representative tasks to be performed in the topographic area include: storage of raw field data; computation, adjustment, plotting and annotation of three dimensional coordinates using coordinate geometry (to include traverse, intersection, and resection); computation of a level line; computation of astronomic azimuth (stellar); coordinate conversion of universal transverse mercator (UTM)

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to geographic, world geodetic system (WGS); state plane coordinate conversion; zone to zone conversion (UTM); and computation of the convergence of coordinate lines. In the construction survey task area, these representative tasks are critical: storage of raw field data; computation, adjustment, plotting and annotation of three dimensional coordinates using coordinate geometry (to include traverse, intersection and resection); computation and adjustment of level data; computation and formatting for field location, road, airfield, buildings, and utilities (centerline, grade, earthwork); computation and plotting of profiles, cross sections, vertical curves, and horizontal curves; computation and plotting of preliminary/final road, airfield and site construction plans. Software shall be interactive, menu-driven, and user friendly to the maximum extent possible. The software shall feature provisions for entry and automatic transfer of data (prism constant, scale correction, coordinate and alignment data, etc.) from/to the AISI. Objects (shapes, lines, etc.) in software shall be (re)definable, or not less than ten distinct sizes shall be provided. The program shall permit panning, to relocate on a plat, and be capable of providing a plan view of the entire site. It is desirable that drawing layers be implemented, and output to .DWG and .DXF formats be afforded.

3.11 Environmental requirements. The AISI shall endure environmental effects as listed below. The Environmental Test Methods and Engineering Guidelines, MIL-STD-810 may be used as reference.

3.11.1 Operational temperature. The AISI, without use of the winterization kit, shall be capable of operations in climatic conditions between -20 and 50 ºC. The winterized AISI shall operate with the accuracy specified herein to -32 ºC.

3.11.2 Storage temperature. The AISI shall not be damaged by storage at temperatures as low as -46 °C and as high as 71°C, and shall operate as specified herein after stabilization (see 7.5.1).

3.11.3 Solar radiation. The AISI shall operate without damage, within the accuracy specified herein, when subjected to solar radiation of 355 Btu/ft^2/hour, for not less than 4 hours in an ambient air temperature of 43 +0, -2 ºC.

3.11.4 Humidity. The AISI shall operate within the accuracy specified herein and shall not be damaged when subjected to relative humidities of not less than 95 percent at air temperatures between -20 ±1, and 50 ±1 ºC.

3.11.5 Rain. The AISI, mounted in the operating position, with an instrument cover in place, shall show no evidence of water penetration after exposure to windblown rainfall of 10 cm/hr, at wind speeds not less than 64 km/hr, for a period of not less than 30 minutes.

.

3.11.6 Vibration. The AISI, packaged in its storage case, shall endure 45 minutes of vibration in each of the 3 axes. The test bed shall produce a 2.5–3.0 cm double-amplitude orbital path at 5 Hz.

3.11.7 Shock. The AISI, packaged in its storage case, shall withstand one drop from 122 cm onto each of the following: each of the 6 faces, each of the 8 corners, and each of the 12 edges, for a total of 26 drops onto 5-6 cm thick plywood backed on concrete.

3.11.8 Blowing dust. The AISI shall endure a blowing dust environment in the storage mode. The small particle procedure (dust and fine sand) shall be conducted in a test chamber to determine the effects of dust and sand on the AISI while in its transit case. The test shall be conducted with the following criteria:

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a. The air velocities used shall be a maintained minimum of 1.5 m/s to a velocity of 8.9 m/s.
b. The dust composition shall be consistent with red china clay or silicon flour.
c. The dust concentration shall be maintained at 10.6±7 g/m3.
d. The blowing dust test shall be conducted for 6 hours at 23 °C, and for 6 hours at 50 °C.
e. The unit shall be oriented so that each surface shall face the blowing dust for an equal amount of time during each period; experiencing gusts to 8.9 m/s for not less than 15 minutes, intermittently.

3.11.9 Blowing sand. The AISI shall endure a blowing sand environment in the storage mode. The large particle procedure shall be conducted in a test chamber to determine the effects of blowing sand on the AISI while in its transit case. The test shall be conducted with the following criteria:

a. The air velocities used shall be 18 to 29 m/s.
b. The suggested sand composition (at least 95% silicate oxide by weight) is as follows:
    1.00% shall be retained by a 20 mesh screen.
    2.00% shall be retained by a 30 mesh screen.
    15.00% shall be retained by a 40 mesh screen.
    37.00% shall be retained by a 50 mesh screen.
    30.00% shall be retained by a 70 mesh screen.
    15.00% shall be retained by a 100 mesh screen.
c. The sand concentration for natural conditions is 2.2 ±0.50 g/m^3.
d. The blowing sand test shall be conducted at 50 °C.
e. The unit shall be oriented so that each surface shall face the blowing sand for not less than 90 minutes per surface; experiencing gusts to 29 m/s for not less than 30 minutes, intermittently.

3.12 Electromagnetic interference. The instrument shall have design characteristics for the control of electromagnetic interference emission and susceptibility. MIL-STD-461 may be used as a reference in implementing controls.

3.12.1 Conducted emissions. Conducted emissions (CE) on power leads and returns, that obtain power from sources not part of the unit shall not exceed the values in figure 1 for a frequency scan from 10 kHz to 10 MHz.

3.12.2 Conducted susceptibility. Conducted susceptibility (CS) on all interconnecting cables, including power cables, shall not exceed the values in figure 2 for a frequency spectrum scan from 10 kHz to 10MHz. Use curve A for frequency ranges 10 kHz to 2MHz, and curve B for frequency ranges from 2MHz to 10MHz.

3.12.3 Radiated emissions. Radiated emissions (RE) from the instrument and all interconnecting cables shall not exceed the values in figure 3 for a frequency spectrum scan from 2 MHz to 18 GHz. Testing is required up to 1 GHz or 10 times the highest intentionally generated frequency within the instrument, which ever is greater. Measurements beyond 18 GHz are not required.

3.12.4 Radiated susceptibility. Radiated susceptibility (RS) shall not exceed 50.00 volts/meter for a frequency spectrum scan from 2 MHz to 18 GHz. The requirement shall be met for vertically polarized fields up to 30 MHz, and for both horizontally and vertically polarized fields above 30 MHz. Circular polarized fields are not acceptable.

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4 REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials (see 7.5.2) to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulations (FAR).

5 QUALITY ASSURANCE PROVISIONS. The supplier shall comply with the following for certification and acceptability.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability. The AISI offered shall be of the latest model of the standard product line used in the field for at least one year by government or commercial organizations; with a demonstrated reliability and established maintenance and logistics programs.

5.2.1 Examination. Each AISI shall be examined for compliance with the requirements specified in 3.2 through 3.12. No lateral motion shall exist between the electro-optics housing and support. Lenses shall be free of scratches, chips, air bubbles between cemented elements, and cloudiness or dirt between glass and coated film. Optics shall show sharpness of focus and distinct images to provide favorable definition. Focusing components or assemblies shall retain their focus when finger pressure is released and the telescope shall not change focus when transited. Metal surfaces on the optical components for prisms shall not shift. There shall be no flickering of the illumination system when the vertical or horizontal motions are rotated. Liquid within the level vials shall be clear, and the vials shall be tight in the housings. Mirrors shall not tarnish, and backing shall be uniform and unspotted. Any modification necessary to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects lessening stated efficiency shall constitute cause for rejection.

6 PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7 NOTES.

7.1 Procurement. When this Commercial Item Description is used for procurement, the product conformance clause (see 5.1) must appear in the solicitation.

7.2 Acquisition requirements. Acquisition documents will specify the following:

a. Title, number, and date of this specification.
b. Type of AISI required (see 2.1).
c. Class of AISI required (see 2.2).
d. Style of AISI required (see 2.3).
e. Quantity of survey system, software packages per AISI (see 3.4.5 and 7.3).
f. When a prism adapter is required (see 3.4.6).
g. Optional equipment required (see 3.4.9).
h. When is separate winterization kit is not required (see 3.4.9.2 and 7.3).
i. Level of preservation and packing required.

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7.3 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, maintenance and operation manuals, and specify the number of survey system software packages to be furnished with each AISI. When procuring a winterized unit that is not provided with a separate kit (see 3.4.9.2), the contracting officer should also secure documentation certifying the ability of the AISI to perform to the requirements of this specification.

7.4 Sources of documents.

7.4.1 ASTM F 1166 is available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

7.4.2 A-A-52298, and MIL-STD-461, are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

7.4.3 TB MED-524 is available from the Commander, U.S. Army Publication Distribution Center, 1655 Woodson Road, St Louis, MO 63114.

7.4.4 Standards and Specifications for Geodetic Control Networks, Sept. 1984, is available from the National Geodetic Information Branch, N/OG174 Rockwall Building, Room 24, National Geodetic Survey (NOAA), Rockville, MD 20852.

7.5 Definitions. The following definitions apply in this document.

7.5.1 Stabilization. Stabilization is a 2 °C, or less, differential between the ambient temperature and that of the instrument.

7.5.2 Recovered materials. For the purpose of this requirement, recovered materials are those that have been collected from solid waste and reprocessed to become a source of raw materials; as distinguished from virgin raw materials. Used, rebuilt, or remanufactured components, pieces, and parts shall not be construed as recovered.

7.6 National Stock Numbers (NSN). The following NSNs have been used in the past to procure style I survey instruments and the tripod base:

Type I	6675-01-372-0047
Type II	6675-01-371-4961
Tripod Base	1290-01-468-6062

7.7 Subject term (key word) listing.

Electronic angle measurement

Electronic distance measurement

One-man survey station

Theodolite

Total station

Transit

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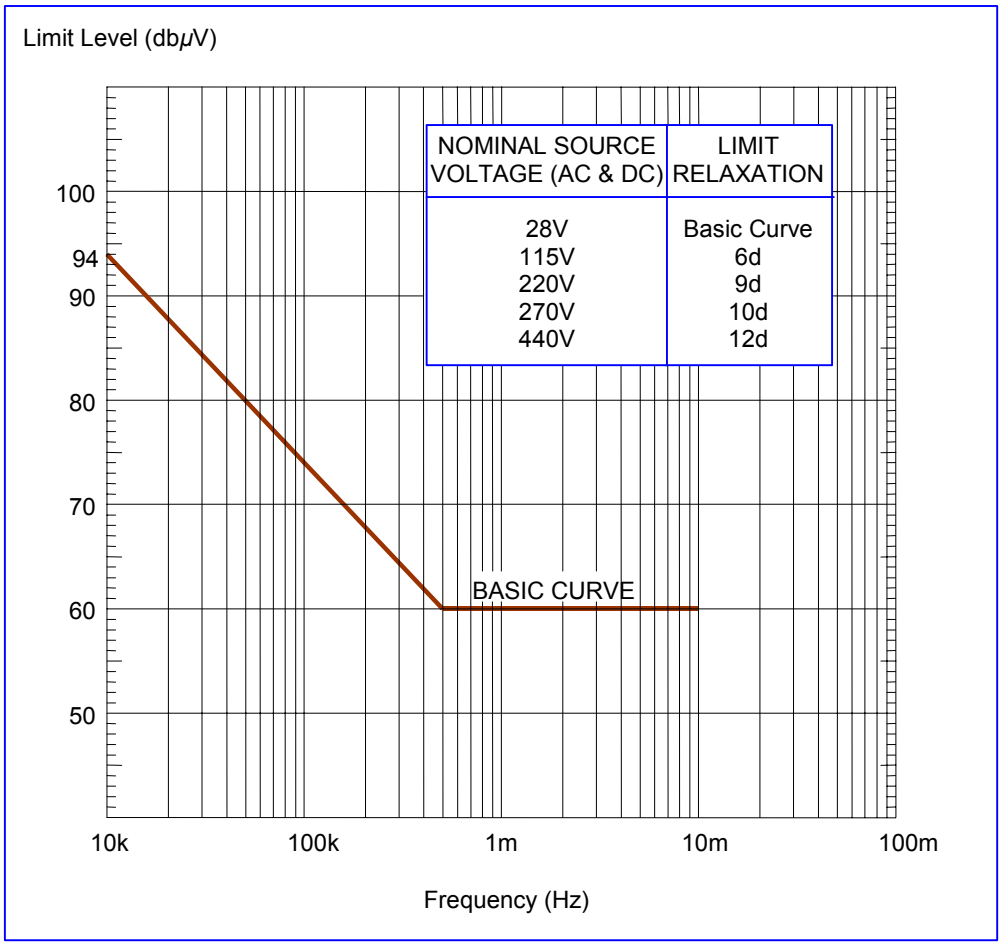


Figure 1. Conducted emission limits.

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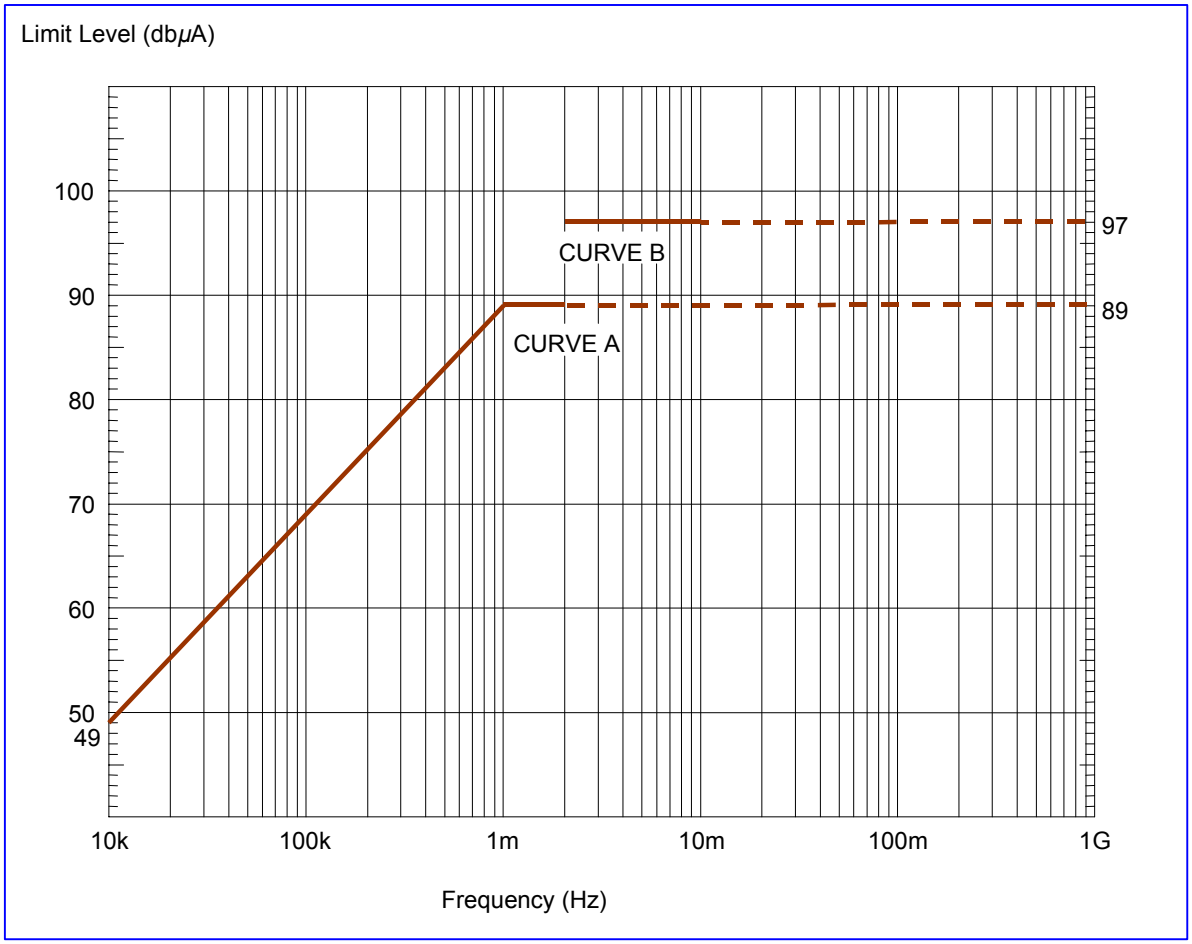


Figure 2. Conducted susceptibility limits.

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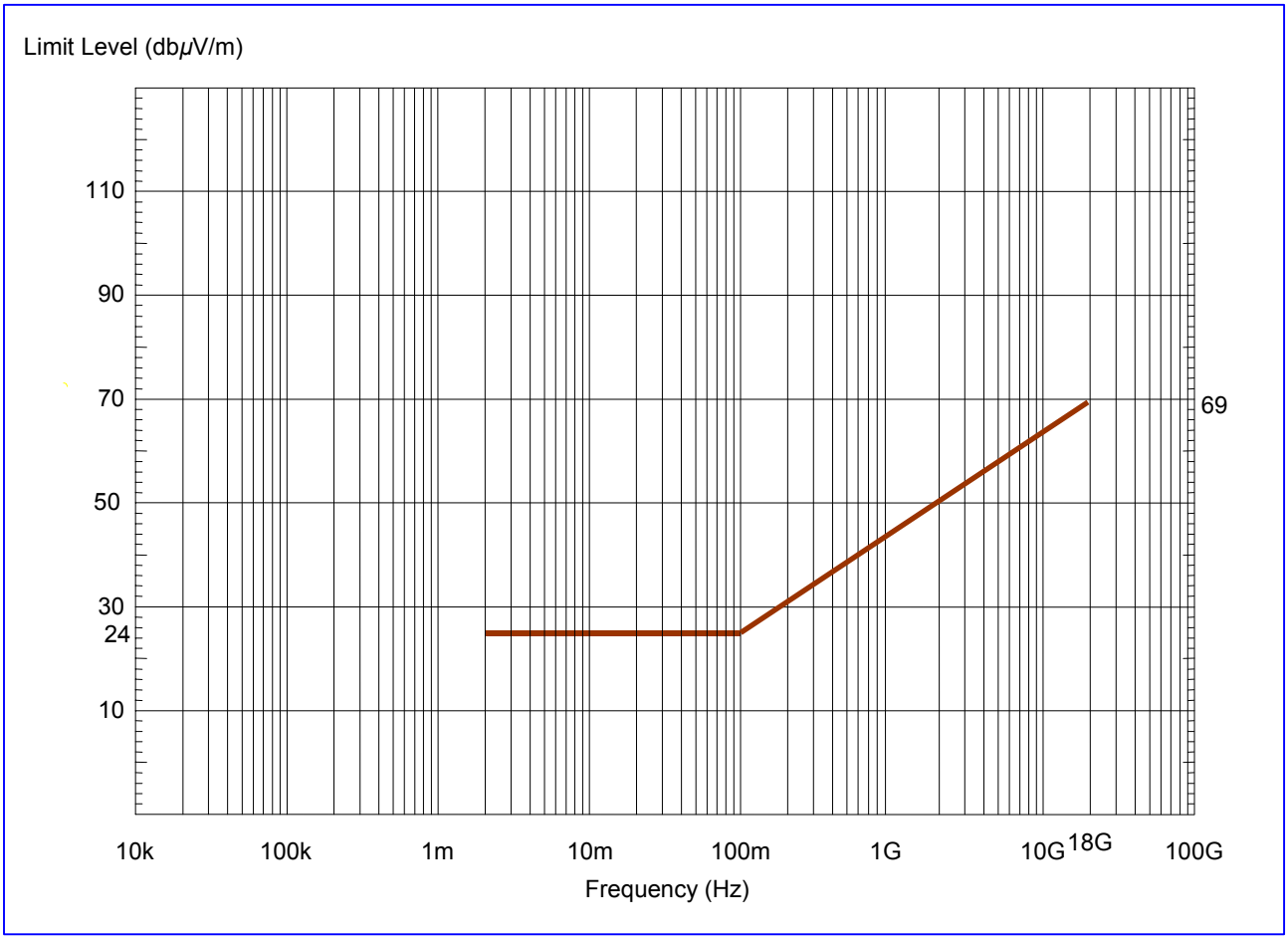


Figure 3. Radiated emission limits.

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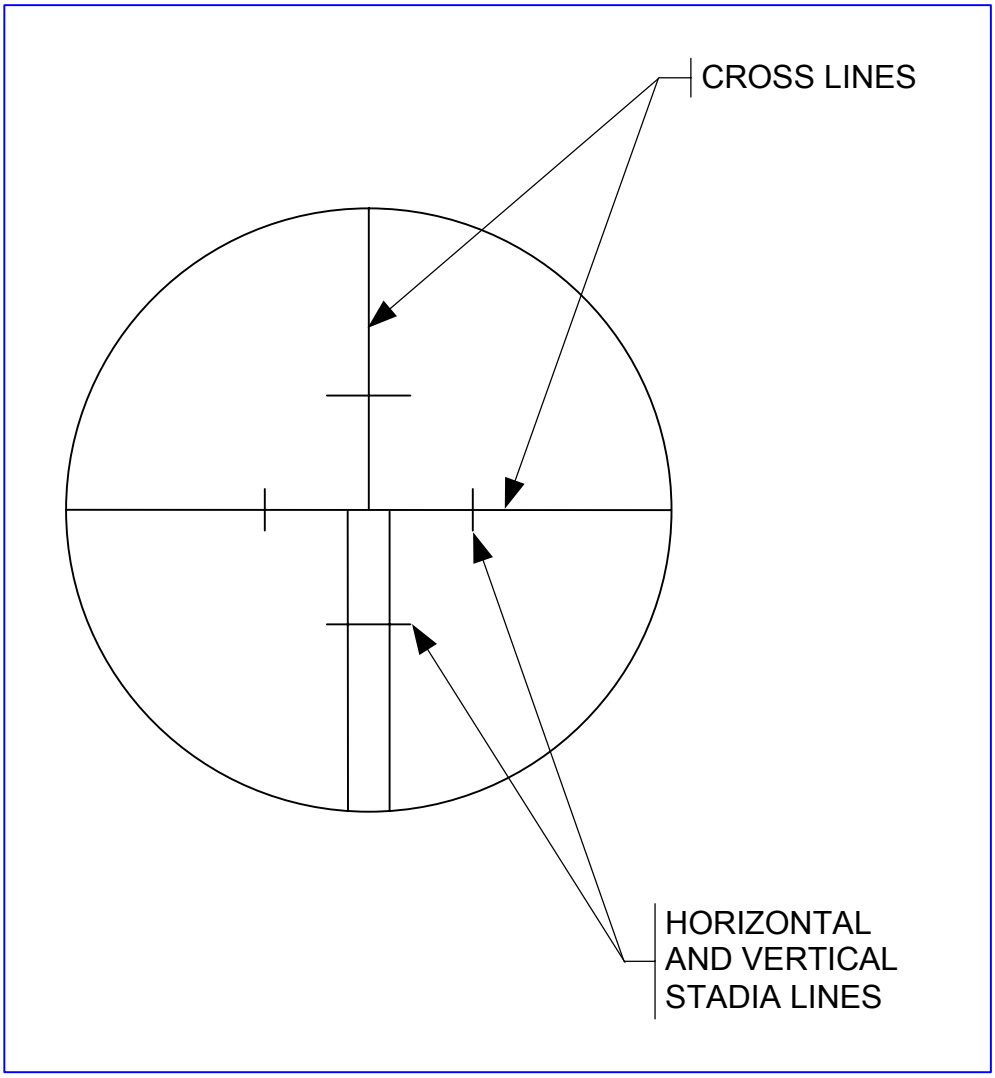


Figure 4. Reticle pattern for theodolite.

(Illustrative example only)

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Custodians:
Army – CE 3
Air Force – 99
Navy – YD

Review Activities:
Army - CR
DLA – GS

Civilian Coordinating Activity:
GSA-FS

Preparing Activity:
Army – CE 3
Project 6675-0155

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-52567A	2. DOCUMENT DATE *(YYYYMMDD)* 20021031

3. DOCUMENT TITLE
SURVEY INSTRUMENT, AUTOMATED, INTEGRATED (AISI)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER	b. TELEPHONE (*Include Area Code*) (1) Commercial: (703) 428-6862 (2) DSN: 328-6862
c. ADDRESS (*Include Zip Code*) ATTN: CEERD-TS-T 7701 TELEGRAPH ROAD ALEXANDRIA, VA 22315-3864	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Standardization Program Office (DLSC-LM)** **8725 John J. Kingman Road, Suite 2533** **Fort Belvoir, Virginia 22060-6221** **Telephone (703) 767-6888 DSN 427-6888**

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